

EDUCATIVE COMMENTARY ON ISI 2023 MATHEMATICS PAPERS

(Last Revisedd 07/06/2023)

Introduction	1
Paper 1	2
Paper 2	29
Concluding Remarks	43

Introduction

The original motivation for writing the annual commentaries on ISI B.Stat. Entrance Tests is now waning. They began in 2016 because it was the year in which some exciting questions came to the notice of the author. Later on commentaries from 2014 were added. Unfortunately, after 2017, there was a noticeable decline in the quality of questions. Some readers also brought to the notice of the author that a few questions which were hailed in the commentary were picked from some reputed international contests. This did not detract from the intrinsic quality of the questions. Nor is it likely that some candidates had seen the solutions already. But apart from loss of originality, it is an insult to a good problem (for which originally ample time was allowed) to expect the candidates to crack it in a couple of minutes.

Still, the annual commentaries on ISI Entrance Test continued more as a mission, because some students pointed out that unlike the JEE, there are no sources to get the answers of ISI Entrance tests. But things are changing here too. In addition to the official key to the multiple choice questions in Paper 1, videos giving worked out solutions are available. As for the second paper, solutions for some of the problems are presented on the [Artofproblemsolving.com](https://artofproblemsolving.com).

But even with both the justifications for the commentaries losing their strengths, I believe there is still some room to bring out the educative aspects of some of the problems and solutions. So, here is the commentary on ISI 2022.

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, Third Edition, published by Universities Press, Hyderabad.

I am thankful to the persons who made the question papers available to me. The pure geometry solution to Q.3 in Paper 2 was contributed by Pankaj Meena while the alternate solution to Q.6 in it was contributed by Saad Hassan. Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2023 Paper 1 with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 For a real number x ,

$$x^3 - 7x + 6 > 0$$

if and only if

- (A) $x > 2$ (B) $-3 < x < 1$
 (C) $x < -3$ or $1 < x < 2$ (D) $-3 < x < 1$ or $x > 2$

Answer and Comments: (D). There is a formula to give the roots of a cubic. But it is far more complicated than the quadratic formula and is seldom used. But when one root of a cubic can be found easily, it can be factored as the product of a linear polynomial and a quadratic polynomial. The roots of the latter are the missing two roots of the cubic. In the present problem, $x^3 - 7x + 6$ has 1 as an obvious root and it factors as $(x - 1)(x^2 + x - 6)$ and further as $(x - 1)(x + 3)(x - 2)$. Thus there are three distinct real roots, viz. $-3, 1$ and 2 in ascending order. The cubic is positive when all three factors are positive, which is the case for $x > 2$, or when two of them are negative and the third positive, for which to happen, x must lie between -3 and 1 . Put together, (D) is the answer.

An absolutely simple problem since all three roots are easy to identify and distinct. The problem would have been more interesting if there were only one real root, or a double root.

Q.2 The number of consecutive zeroes adjacent to the digit in the units place of 401^{50} is

- (A) 3 (B) 4 (C) 49 (D) 50

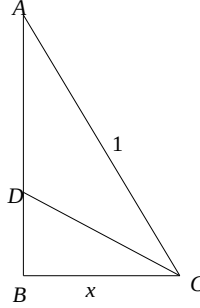
Answer and Comments: (A). Since $401 = 400 + 1$, we can expand 401^{50} by the binomial theorem. The last term would be 1. The question asks to identify the terms before this and to count the number of trailing zeros in their sum.

It is a little easier to rewrite $(401)^{50}$ as $(1 + 400)^{50}$ and find the numbers of trailing zeros from the first terms onwards. The first term, viz. 1 has no trailing zeros. The second term is $\binom{50}{1}(400)^1 1^{49}$, i.e. 20000. When added to 1, we get 20001 which has three zeros adjacent to 1. The remaining terms will have higher powers of 100 dividing them. So when added to 20001, the tail end of the sum will remain 20001. So the answer is 3.

A simple but good problem on binomial expansion. Although the reasoning is time consuming to write out, it does not take much time to

conceive it and then the calculations are minimal. So the problem suits well as an MCQ.

- Q.3 Consider a right angled triangle $\triangle ABC$ whose hypotenuse AC is of length 1. The bisector of $\angle ACB$ intersects AB at D . If BC is of length x , then what is the length of CD ?



- (A) $\sqrt{\frac{2x^2}{1+x}}$ (B) $\frac{1}{\sqrt{2+2x}}$
 (C) $\sqrt{\frac{x}{1+x}}$ (B) $\frac{1}{\sqrt{1-x^2}}$

Answer and Comments: (A). Since $CD^2 = BC^2 + BD^2 = x^2 + BD^2$, we shall be through if we can find BD . By the angle bisector property the ratio $BD : DA$ equals $x : 1$. But $BD + DA = BA = \sqrt{1-x^2}$. Hence $BD = \sqrt{1-x^2} \times \frac{x}{x+1}$. So $CD^2 = x^2 + BD^2 = x^2 + (1-x^2) \times \frac{x^2}{(x+1)^2}$, which simplifies to $\frac{2x^2}{1+x}$. Hence (A) is the answer.

A simple trigonometric problem based on the Pythagoras theorem and the angle bisection property. The calculations are somewhat prone to mistakes of hand.

- Q.4 Define a polynomial $f(x)$ by

$$f(x) = \begin{vmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{vmatrix}$$

for all $x \in \mathbb{R}$, where the right hand side is the determinant. Then the roots of $f(x)$ are of the form

- (A) $\alpha, \beta \pm i\gamma$ where $\alpha, \beta, \gamma \in \mathbb{R}$, $\gamma \neq 0$ and i is a square root of -1 .
 (B) α, α, β where $\alpha, \beta \in \mathbb{R}$ are distinct.
 (C) α, β, γ where $\alpha, \beta, \gamma \in \mathbb{R}$ are all distinct.
 (D) α, α, α for some $\alpha \in \mathbb{R}$.

Answer and Comments: (B). The notation for the determinant is so standard that it was hardly necessary to specify what it meant. Perhaps it is an implied hint that some properties of the determinant are to be used. If we expand the determinant it will come out to be a cubic in x . The problem then will be similar in spirit to Q.1 which also dealt with the roots of a cubic. In the present problem, if we look at the determinant, we see easily that it vanishes for $x = 1$ as two rows become identical. Hence $x - 1$ is a factor of the determinant, or equivalently, 1 is a root of the cubic $f(x)$. Also, if you add the second and the third rows to the first, you get $2x + 1$ as a factor of $f(x)$. Hence $x = -\frac{1}{2}$ is also a root.

What about the third root? We can expand $f(x)$ and find it after factoring $(x - 1)(2x + 1)$ out. But there is a more intelligent way. If we are extra sharp we will notice that $(x - 1)$ will be a double factor, because putting $x = 1$ makes the first two rows equal and this is enough for the determinant to vanish. But putting $x = 1$ also makes the second and the third rows equal, *independently of the equality of the first two rows* and hence $x - 1$ is a factor again. (We cannot apply this reasoning to the first and the third row now because their equality follows from that of the first two rows and of the last two rows. So it cannot give us anything more. In particular, we cannot conclude that $(x - 1)^3$ is a factor of $f(x)$.) In fact, once we know that $(x - 1)^2$ and $2x + 1$ are factors of the determinant, there is no need to expand it. Since it is a polynomial of degree 3 in x , the other factor, if any, must be a constant and its presence does not affect the roots. As 1 is a double root and $-\frac{1}{2}$ is the third root, the correct answer is (B).

The problem is a good way to reward those who work it without expansion. But the MCQ nature of the question mars its beauty because as it happens, in the present case a brute force expansion of the determinant and further factorisation is not prohibitively lengthy. The problem is actually a part of the JEE 2001 question discussed at the end of Comment No. 6 of Chapter 24. That problem asked to find the number of distinct roots of

the equation
$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$
 lying in the interval $[-\pi/4, \pi/4]$.

Q.5 Let S be the set of those real numbers x for which the identity

$$\sum_{n=2}^{\infty} \cos^n x = (1 + \cos x) \cot^2 x$$

is valid, and the quantities on both the sides are finite. Then

- (A) S is the empty set.
- (B) $S = \{x \in \mathbb{R} : x \neq n\pi \text{ for all } n \in \mathbb{Z}\}$.
- (C) $S = \{x \in \mathbb{R} : x \neq 2n\pi \text{ for all } n \in \mathbb{Z}\}$.
- (D) $S = \{x \in \mathbb{R} : x \neq (2n + 1)\pi \text{ for all } n \in \mathbb{Z}\}$.

Answer and Comments: (B). The L.H.S. is an infinite geometric series with starting term $\cos^2 x$ and common ratio $\cos x$. So it will be finite if and only if $-1 < \cos x < 1$. So real numbers of the form $n\pi$, for $n \in \mathbb{Z}$ are to be excluded from S . To see if there are any additional restrictions, we solve the equation for the values of x other than those of the form $n\pi$. For such values, the L.H.S. sums to $\frac{\cos^2 x}{1 - \cos x}$. Hence the equation becomes

$$\frac{\cos^2 x}{1 - \cos x} = (1 + \cos x) \cot^2 x \quad (1)$$

which simplifies to

$$\begin{aligned} \cos^2 x &= (1 - \cos^2 x) \frac{\cos^2 x}{\sin^2 x} \\ &= \cos^2 x \end{aligned} \quad (2)$$

which is true for all x for which $\sin x \neq 0$. But that happens only when x is an integral multiple of π , which is already excluded. Hence no new restrictions are imposed by the equation. Therefore the solution set S is $\{x \in \mathbb{R} : x \neq n\pi, n \in \mathbb{Z}\}$.

An absolutely simple problem.

Q.6 Let

$$S = \left\{ \left(\theta \sin \frac{\pi\theta}{1+\theta}, \frac{1}{\theta} \cos \frac{\pi\theta}{1+\theta} \right) : \theta \in \mathbb{R}, \theta > 0 \right\}$$

and $T = \{(x, y) : x, y \in \mathbb{R}, xy = \frac{1}{2}\}$. How many elements does $S \cap T$ have?

- (A) 0 (B) 1 (C) 2 (D) 3

Answer and Comments: (B). The set T is a familiar hyperbola with asymptotes along the axes. This is of little help here since the question merely asks how many points (x, y) on this hyperbola lie in the set S . For this to happen, x and y must be, respectively, of the form $\theta \sin \frac{\pi\theta}{1+\theta}$ and $\frac{1}{\theta} \cos \frac{\pi\theta}{1+\theta}$ for the same θ . But then xy would equal $\sin \frac{\pi\theta}{1+\theta} \cos \frac{\pi\theta}{1+\theta}$ which is precisely $\frac{1}{2} \sin \frac{2\pi\theta}{1+\theta}$. Therefore the condition $xy = \frac{1}{2}$ reduces to $\sin \frac{2\pi\theta}{1+\theta} = 1$. If we solve this indiscriminately, we would get

$$2\pi\theta = (1 + \theta)(2\pi k + \frac{\pi}{2}) \quad (1)$$

for some integer k . For each integer k , there will be a solution for θ which can be written down, after some simplification, as

$$\theta = \frac{4k + 1}{3 - 4k} \quad (2)$$

It is thus tempting to think that $S \cap T$ is an infinite set. As this is not given as an alternative, we need to check further. Note that θ is required to be positive. This means the numerator and the denominator of the ratio on the R.H.S. of (2) must be of the same sign. As k takes only integral values, if $k > 0$ then the numerator is positive and the denominator negative. The opposite holds if k is a negative integer. So, it is only for $k = 0$, that $\theta > 0$. Hence there is only one point in $S \cap T$. (Although not asked, this common point occurs for $\theta = \frac{1}{3}$. Hence it is the point $(\frac{1}{3\sqrt{2}}, \frac{3}{\sqrt{2}})$.)

An alternate method to rule out the multiplicity of common points of $S \cap T$ is to observe that for $\theta > 0$, the ratio $\frac{\theta}{1+\theta}$ lies between 0 and 1. (It increases monotonically from 0 to 1 as θ varies over $(0, \infty)$.) Hence $\frac{2\pi\theta}{1+\theta}$ lies in the interval $(0, 2\pi)$ for all $\theta > 0$ and there is only one point in this interval, viz. $\frac{\pi}{2}$ at which the sine function takes the value 1.

The problem requires some care in eliminating other candidates of $S \cap T$. By not giving ‘infinitely many’ as a possible answer, the paper-setters have pointed out the need for such care.

Q.7 How many numbers formed by rearranging the digits of 234578 are divisible by 55?

- (A) 0 (B) 12 (C) 36 (D) 72

Answer and Comments: (B). Since 11×5 is the prime factorisation of 55, divisibility by 55 is equivalent to simultaneous divisibility by 5 and by 11. The first is easy to check, since a number is divisible by 5 if and only if its last digit is 5 or 0. In the present case, this means that the number is of the form $abcde5$ where $abcde$ is a permutation of 23478. There are $5! = 120$ such permutations and we have to find those which, after putting a 5 at the end give a number divisible by 11.

The criterion for divisibility by 11 is that the sums of the alternate digits should differ by a multiple of 11. In the present problem, this means that the difference $(a + c + e) - (b + d + 5)$ should be a multiple of 11. Since $a + c + e + b + d + 5$ equals the sum of the original digits 2, 3, 4, 5, 7, 8, viz. 29, and is odd, the difference has to be odd too. Moreover it cannot exceed 29. So divisibility by 11 is possible only in the two cases $(a + c + e) - (b + d + 5) = \pm 11$. The first possibility gives $a + c + e = 20$ and $b + d + 5 = 9$, the second gives $a + c + e = 9$ and $b + d + 5 = 20$. The first possibility cannot occur because no three digits out of 2, 3, 4, 7, 8 can make a total of 20. In the second possibility we have $b + d + f = 20$, i.e. $b + d = 15$ which is possible only when $b = 7, d = 8$ or vice versa. In each case, the remaining three digits, viz. 2, 3, 4 have to add up to 9. They can be put as a, c, e in $3! = 6$ ways. Hence in all there are 6×2 numbers of the given type that are divisible by 55.

A simple but good problem requiring very elementary number theory and combinatorics. The reasoning appears long when written down fully.

But it does not take much time to do it as a rough work. So the problem is well suited as a MCQ.

Q.8 The limit

$$\lim_{n \rightarrow \infty} n^{-\frac{3}{2}} \left((n+1)^{(n+1)} (n+2)^{(n+2)} \dots (2n)^{(2n)} \right)^{\frac{1}{n^2}}$$

equals

$$(A) 0 \quad (B) 1 \quad (C) e^{-\frac{1}{4}} \quad (D) 4e^{-\frac{3}{4}}$$

Answer and Comments: (D). Call the given expression as a_n . It is a complicated expression involving, besides powers, the product of n factors. As the number of factors changes with n , theorems such as the limit of a product is the product of limits are not applicable.

The standard practice in such cases (after ensuring that it is positive for all n) is to take the logarithm, say b_n , of the given expression, find its limit, say L , and then take e^L as the limit of the original sequence. So we set $b_n = \ln a_n$ to get, using properties of logarithms,

$$\begin{aligned} b_n = \ln a_n &= -\frac{3}{2} \ln n + \frac{1}{n^2} \sum_{k=1}^n (n+k) \ln(n+k) \\ &= -\frac{3}{2} \ln n + \frac{1}{n} \sum_{k=1}^n \left(1 + \frac{k}{n}\right) \ln(n+k) \end{aligned} \quad (1)$$

If the factor $\ln(n+k)$ were absent in the k -th term of the summation, the second term on the R.H.S. of (1) would be a Riemann sum of the integral $\int_1^2 x dx$, specifically a Riemann sum arising from a partition of the interval $[1, 2]$ into n equal parts. As n tends to infinity, the mesh $\frac{1}{n}$ of the partition tends to 0 and all such Riemann sums tend to the respective definite integrals.

Here the factor $\ln(n+k)$ creates a problem. To tackle it, we rewrite $n+k$ as $(1 + \frac{k}{n})n$. Then $\ln(n+k) = \ln(1 + \frac{k}{n}) + \ln n$ and (1) becomes

$$\ln a_n = -\frac{3}{2} \ln n + \frac{1}{n} \left[\sum_{k=1}^n \left(1 + \frac{k}{n}\right) \ln\left(1 + \frac{k}{n}\right) \right] + \frac{1}{n} \sum_{k=1}^n \left[\left(1 + \frac{k}{n}\right) \ln n\right] \quad (2)$$

which simplifies to

$$\begin{aligned} \ln a_n &= -\frac{3}{2} \ln n + \frac{1}{n} \left[\sum_{k=1}^n \left(1 + \frac{k}{n}\right) \ln\left(1 + \frac{k}{n}\right) \right] + \frac{\ln n}{n} \left[n + \frac{n(n+1)}{2n} \right] \\ &= -\frac{3}{2} \ln n + \frac{1}{n} \left[\sum_{k=1}^n \left(1 + \frac{k}{n}\right) \ln\left(1 + \frac{k}{n}\right) \right] + \ln n \left[\frac{(3n+1)}{2n} \right] \end{aligned} \quad (3)$$

As $n \rightarrow \infty$, the middle term tends to the definite integral $\int_1^2 x \ln x dx$, which we shall evaluate a little later. The first term tends to $-\infty$ while the last one tends to ∞ . But their sum, say c_n , may tend to some finite limit. To find it we note that

$$\begin{aligned} c_n &= \ln n \left(\frac{3n+1}{2n} - \frac{3}{2} \right) \\ &= \frac{\ln n}{2n} \end{aligned} \quad (4)$$

Thus both the numerator and the denominator of c_n tend to ∞ as $n \rightarrow \infty$. But it is quite well known that $\ln n$ tends to ∞ much slower than n and hence the ratio tends to 0. (This can also be seen by replacing n by some continuous variable u which tends to ∞ and applying the l'Hôpital rule.)

Summing up, $\lim_{n \rightarrow \infty} a_n$ is the definite integral $\int_1^2 x \ln x dx$. If we integrate by parts, this comes out to be

$$\begin{aligned} \int_1^2 x \ln x dx &= \left. \frac{x^2}{2} \ln x \right|_1^2 - \int_1^2 \frac{x}{2} dx \\ &= 2 \ln 2 - \frac{3}{4} = \ln 4 - \frac{3}{4} \end{aligned} \quad (5)$$

Therefore the original limit $\lim_{n \rightarrow \infty} a_n$ equals $e^{\ln 4 - \frac{3}{4}} = \frac{4}{e^{\frac{3}{4}}}$.

A complicated and lengthy question involving properties of logarithms, their rates of growth, and finally, the evaluation of a certain definite integral involving logarithms. A better question would have been to do away with the last part, give the expression

$$n^r \left((n+1)^{(n+1)} (n+2)^{(n+2)} \dots (2n)^{(2n)} \right)^{\frac{1}{n^2}}$$

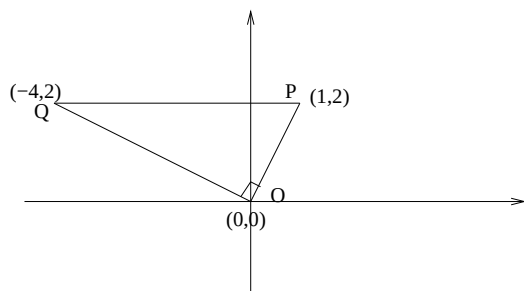
and ask to find for which value of r it has a finite, non-zero limit as $n \rightarrow \infty$. Or, ask which power of n has comparable rate of growth as the expression

$$\left((n+1)^{(n+1)} (n+2)^{(n+2)} \dots (2n)^{(2n)} \right)^{\frac{1}{n^2}}.$$

Q.9 Consider a triangle with vertices $(0, 0)$, $(1, 2)$ and $(4, 2)$. Let A be the area of the triangle and B be the area of the circumcircle of the triangle. Then $\frac{B}{A}$ equals

(A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{4}$ (C) $\frac{3}{\sqrt{2}}\pi$ (D) 2π

Answer and Comments: (B).



Call the vertices as O, P, Q respectively. From the area formula in terms of coordinates of the vertices,

$$\begin{aligned} A &= \pm \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 1 & 2 & 1 \\ -4 & 2 & 1 \end{vmatrix} \\ &= \pm 5 \end{aligned} \quad (1)$$

As the area is positive, we take $A = 5$ sq. units.

There is no similar formula for B , the area of the circumcircle. We would be through if we can identify its radius. But, again, there is no general formula for it either. But we can make use of special features of $\triangle OPQ$. As the slopes of the sides OP and OQ are 2 and $-\frac{1}{2}$ respectively, we get that $\angle POQ$ is a right angle. Hence PQ is a diameter. Its length is $1 - (-4) = 5$ units. Hence the radius of the circumcircle is $\frac{5}{2}$. Therefore, B , its area, is $\frac{25\pi}{4}$ sq. units. Therefore the ratio $\frac{B}{A}$ equals $\frac{5\pi}{4}$.

An absolutely simple problem once it is realised that the triangle is right angled.

Q.10 The value of

$$\sum_{k=0}^{202} (-1)^k \binom{202}{k} \cos \frac{k\pi}{3}$$

equals

(A) $\sin\left(\frac{202}{3}\pi\right)$ (B) $-\sin\left(\frac{202}{3}\pi\right)$ (C) $\cos\left(\frac{202}{3}\pi\right)$ (D) $\cos^{202}\left(\frac{\pi}{3}\right)$

Answer and Comments: (C). Call the given sum as S . If we can find some number t such that $\cos \frac{k\pi}{3} = t^k$ for all integers $k = 0, 1, 2, \dots, 202$, then by the binomial theorem we would have

$$S = (1 - t)^{202} \quad (1)$$

Unfortunately, there is no real number (not even a complex number) t with this property, for if there were then $\{\cos \frac{k\pi}{3}\}$ would be a geometric progression, which it surely is not since $\cos^2 \frac{2\pi}{3} = \frac{1}{4}$, but $\cos \frac{\pi}{3} \cos \frac{3\pi}{3} = -\frac{1}{4}$.

Nevertheless we do have a *complex* number t such that $\cos \frac{k\pi}{3}$ is the *real part* of the k -th power of t . Indeed, if we take

$$t = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \quad (2)$$

then by DeMoivre's theorem we get

$$t^k = \cos \frac{k\pi}{3} + i \sin \frac{k\pi}{3} \quad (3)$$

A compact notation for t is $e^{i\pi/3}$. With this notation, we get that S is the real part of

$$(1 - e^{i\pi/3})^{202} \quad (4)$$

This is not of much use by itself, because if we expand it using the binomial theorem and take its real part, we shall get back to the sum we started with. So let us see if the number $1 - e^{i\pi/3}$ can be recast so that its 202-th power can be calculated efficiently. This turns out to be easy. Indeed

$$\begin{aligned} 1 - e^{i\pi/3} &= 1 - \cos(\pi/3) - i \sin(\pi/3) \\ &= 1 - \frac{1}{2} - i \frac{\sqrt{3}}{2} \\ &= \frac{1}{2} - i \frac{\sqrt{3}}{2} \\ &= \cos(\pi/3) - i \sin(\pi/3) \\ &= e^{-i\pi/3} \end{aligned} \quad (5)$$

Combining this with (4), we now get that S is the real part of

$$(e^{-i\pi/3})^{202} = e^{-i202\pi/3} = \cos \frac{202\pi}{3} - i \sin \frac{202\pi}{3} \quad (6)$$

Hence the given sum S equals $\cos \left(\frac{202\pi}{3} \right)$.

A good problem based on the elementary properties of complex roots of unity.

Q.11 For real numbers $a, b, c, d, a', b', c', d'$, consider the system of equations

$$\begin{aligned} ax^2 + ay^2 + bx + cy + d &= 0 \\ a'x^2 + a'y^2 + b'x + c'y + d' &= 0 \end{aligned}$$

If S denotes the set of all real solutions (x, y) of the above system of equations, then the number of elements in S can never be

- (A) 0 (B) 1 (C) 2 (D) 3

Answer and Comments: (D). Clearly $S = S_1 \cap S_2$ where S_1 is the solution set of the first equation and S_2 that of the second. Assuming a and a' are non-zero, both S_1 and S_2 are circles since neither equation has a xy -term. Now, two circles may have no point in common (i.e. be disjoint), or just one point in common (when they touch each other) or two points in common. But if they have 3 points in common, then they are identical and hence have infinitely many points in common. So the number of points of intersection can never be 3.

The degenerate cases when a and/or a' vanishes are easy to deal with. For then S_1 and/or S_2 is a straight line. A circle cannot meet a straight line in 3 (distinct) points, nor can two straight lines. (Those who are very fussy, will also consider the cases where these lines are degenerate. But even then $S_1 \cap S_2$ cannot have exactly 3 points.

The underlying theme of the problem is that a circle is uniquely determined by three points. Once this is realised, there is no computation involved. This problem, is therefore, a good test of the ability to get down to the underlying theme. Unfortunately, the data does not specify that the equations are non-degenerate. Those who consider the degeneracies are wasting their time. Yet another illustration where scruples do not pay in an MCQ.

Q.12 Let f, g be continuous functions from $[0, \infty)$ to itself,

$$h(x) = \int_{2^x}^{3^x} f(t)dt, \quad x > 0,$$

And

$$F(x) = \int_0^{h(x)} g(t)dt, \quad x > 0$$

If F' is the derivative of F , then for $x > 0$,

- (A) $F'(x) = g(h(x))$
- (B) $F'(x) = g(h(x))[f(3^x) - f(2^x)]$
- (C) $F'(x) = g(h(x))[x3^{x-1}f(3^x) - x2^{x-1}f(2^x)]$
- (D) $F'(x) = g(h(x))[3^x f(3^x) \ln 3 - 2^x f(2^x) \ln 2]$

Answer and Comments: (D). The theme of the problem is the derivative of a function defined by an integral. The first equation defines $h(x)$ as the integral of the function $f(x)$ whose upper and lower limits are both functions of x . Using a generalisation of the Fundamental Theorem of Calculus, we have

$$\begin{aligned} h'(x) &= f(3^x) \frac{d}{dx}(3^x) - f(2^x) \frac{d}{dx}(2^x) \\ &= 3^x \ln 3 f(3^x) - 2^x \ln 2 f(2^x) \end{aligned} \quad (1)$$

The second equation defines a new function $F(x)$ by integrating yet another continuous function $g(x)$ (about which nothing is given) in the limits 0 to $h(x)$. As the lower limit is constant, we have

$$F'(x) = g(h(x))h'(x) \quad (2)$$

If we substitute (1) into (2), we get (D).

An absolutely simple, in fact dull, problem based on differentiation of a function defined by an integral.

- Q.13 Suppose x and y are positive integers. If $4x + 3y$ and $2x + 4y$ are divided by 7, then the respective remainders are 2 and 5. If $11x + 5y$ is divided by 7, then the remainder equals

(A) 0 (B) 1 (C) 2 (D) 3

Answer and Comments: (B). The given statements can be written down as

$$4x + 3y = 7p + 2 \quad (1)$$

$$\text{and } 2x + 4y = 7q + 5 \quad (2)$$

where p, q are some integers. As their values are not given, these equations cannot be solved for x and y . Nor is it needed because the question merely asks the remainder when $11x + 5y$ is divided by 7. We could change x and y by adding any multiples of 7 and that will not affect the remainders.

It is therefore convenient to work in terms of the residue classes modulo 7. (See Comment No. 15 of Chapter 4. Regrettably, the Index gives a wrong location.) The residue class modulo 7 of an integer a is generally denoted by $[a]_7$ or simply by $[a]$ when the modulus 7 is understood. The basic rules of residue arithmetic are $[a \pm b] = [a] \pm [b]$ and $[ab] = [a][b]$. As a result, the two equations above reduce to

$$[4][x] + [3][y] = [2] \quad (3)$$

$$\text{and } [2][x] + [4][y] = [5] \quad (4)$$

respectively. If we add these two equations, the coefficient of $[y]$ is $[3] + [4] = [7]$ which is the same as $[0]$ modulo 7. Similarly, the right hand sides $[2]$ and $[5]$ add to $[0]$. So we are left with

$$[6][x] = [0] \quad (5)$$

This means that the integer $6x$ is divisible by 7. But as 7 is relatively prime to 6, we must have that 7 divides x , or equivalently, $[x] = [0]$.

The equation (3) now reduces to $[3y] = [2]$. That means that $3y - 2$ is divisible by 7. Again, this does not define y uniquely, but only modulo 7. With trial, we can take y as 3 since $3 \times 3 = 9$ has the same residue class

as 2 modulo 7. (This also tallies with (4) because $4 \times 3 = 12$ is indeed congruent to 5 modulo 7.)

Summing up, we now have that $[x] = [0]$ and $[y] = [3]$. So $[11x + 5y] = [11][0] + [5][y] = [15] = [1]$. Hence when $11x + 5y$ is divided by 7 the remainder will be 1.

A very good problem based on residue arithmetic. Those totally unfamiliar with it are likely to be confused that *prima facie* the data is insufficient. It is only after realising that what is asked is only the remainder that the confusion is resolved.

Q.14 Suppose $F : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function which has exactly one local maximum. Then which of the following is true?

- (A) F cannot have a local minimum.
- (B) F must have exactly one local minimum
- (C) F must have at least two local minima
- (D) F must have either a global maximum or a local minimum

Answer and Comments: (D). As the domain $\mathbb{R}$ is the entire real line, there is no room for controversies that may arise because of end points. Easy counterexamples can be given to dismiss the first two options. For example, take the cosine function. It has infinitely many local maxima and local minima. The value at any local maximum is 1. Take any two consecutive maxima, say 0 and 2π . Modify the the graph of the function on $[0, 2\pi]$ by joining $(0, 1)$ and $(2\pi, 1)$ to the point $(\pi, 2)$. Leave intact the other part of the graph. Then there is a unique local maximum at π . But (A) and (B) fail as there still are infinitely many local minima.

For (C), define $F(x) = \frac{1}{|x|}$ for $|x| \geq 1$. But on $[-1, 1]$ join $(-1, 1)$ and $(1, 1)$ to $(0, 2)$ to get the graph of a function which has a unique local maximum at 0, but no local minimum.

As only one alternative is correct, it has to be (D). For an honest answer, here is a proof. Assume that the unique local maximum of F is at x_0 . If it is not the global maximum, there is some x_1 such that $F(x_1) > F(x_0)$. Without loss of generality assume that $x_1 > x_0$. Then the point, say x_2 where F attains its minimum on the interval $[x_0, x_1]$ will be an interior point of $[x_0, x_1]$ and hence a point of local minimum of F on $\mathbb{R}$.

A purely thought oriented problem. Regrettably, being an MCQ, it is impossible to tell whether those who select (D) can give a proof.

Q.15 Suppose $z \in \mathbb{C}$ is such that the imaginary part of z is non-zero and $z^{25} = 1$. Then

$$\sum_{k=0}^{2023} z^k$$

equals

$$(A) 0 \quad (B) 1 \quad (C) -1 - z^{24} \quad (D) -z^{24}$$

Answer and Comments: (D). Since $z^{25} = 1$, z is one of the twentyfive 25-th complex roots of unity. As 25 is odd, all of them except 1 have non-zero imaginary parts. Moreover, for any such root (other than 1), its powers recur in cycles of length 25 (or sometimes a of length 5, which is the only divisor of 25). In either case, the sum of powers occurring in any cycle is 0.

In this problem, the given sum, say S , consists of 2024 terms $z^0 (= 1), z, z^2, \dots, z^{2023}$. The first 2000 terms from 1 to z^{1999} fall into 80 cycles of 25 consecutive powers each. As each cycle adds to 0, we have

$$S = z^{2000} + z^{2001} + \dots + z^{2022} + z^{2023} \quad (1)$$

There 24 terms equal, respectively, $1, z, z^2, \dots, z^{23}$. If we add z^{24} we get a cycle of 25 consecutive powers. As their sum is 0, we get

$$S = 0 - z^{24} = -z^{24} \quad (2)$$

An easy problem based on the peiodicity of the complex roots of unity of a given order.

Q.16 The limit

$$\lim_{x \rightarrow 0} \left(\cos(x) + \cos\left(\frac{1}{x}\right) - \cos(x) \cos\left(\frac{1}{x}\right) - 1 \right)$$

- (A) equals 0 (B) equals $\frac{1}{2}$
 (C) equals 1 (D) does not exist

Answer and Comments: (A). Neither the second nor the third term has a limit as $x \rightarrow 0$ because $\cos(\frac{1}{x})$ tends to no definite limit. But as in Q.8, sometimes the sum of two functions neither of which has a limit can have a limit. To see if that is so here, we note the function, say $f(x)$ whose limit is to be taken factors as

$$f(x) = (\cos x - 1)(1 - \cos \frac{1}{x}) \quad (0)$$

The first factor tends to 0 as $x \rightarrow 0$. Although the second factor does not tend to any limit, it is bounded. That is, there is some number $B > 0$ such that

$$|1 - \cos \frac{1}{x}| \leq B \quad (1)$$

for all $x \neq 0$. Hence we have

$$|f(x)| \leq B|\cos x - 1| \quad (2)$$

As the second factor of the R.H.S. can be made arbitrarily small for x sufficiently small, so can $|f(x)|$. Therefore $|f(x)| \rightarrow 0$ and hence finally $f(x) \rightarrow 0$ as $x \rightarrow 0$.

A simple problem illustrating that the product of two functions of which one is bounded and the other tends to 0 also tends to 0. The problem would have been a little more interesting if $f(x)$ had an additional factor $\frac{1}{x}$. This factor is not bounded near $x = 0$. But $\cos x - 1$ is of the order of x^2 and so $\frac{\cos x - 1}{x}$ tends to 0 too. (Of course, this can also be seen from the fact that $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x}$ is the derivative of $\cos x$ at $x = 0$.)

We pointed to Q.8 where the sum of two sequences had a limit when neither did. So, in a sense their ‘bad’ parts canceled each other, rather like two jagged pieces of a jigsaw puzzle fitting together to form something smooth. In the present problem, the sum of the two terms not having a limit did not have a limit either. But between two factors, if the one without a limit is bounded and the other tends to 0, then the product tends to 0. The vanishing factor by its strength ‘cures’ the anomaly. The reasoning will not apply if the other factor had a limit different from 0.

Q.17 Let n be a positive integer having 27 divisors including 1 and n , which are denoted by $d_1, \dots, d_{27}$. Then the product of $d_1, d_2, \dots, d_{27}$ equals

$$(A) \ n^{13} \quad (B) \ n^{14} \quad (C) \ n^{\frac{27}{2}} \quad (D) \ 27n$$

Answer and Comments: (C). In general, whenever d is a divisor of an integer n so is $\frac{n}{d}$. They occur in pairs and their product equals n . So the smaller one among them is less than $\sqrt{n}$ and the other is bigger. This also means that the number of factors of n is even. The only exception is when n is a perfect square, say $n = k^2$. In this case the factor k is counted only once. All others pair off. So the number of factors is odd.

The present problem falls in this case. As the number of factors, viz. 27, is odd, n is a perfect square and $n^{1/2}$ is an integer. The remaining 26 factors pair off into 13 pairs and each pair multiplies to n . Hence the product of all 27 factors of n is $n^{13}n^{1/2} = n^{\frac{27}{2}}$.

An absolutely simple problem once the key idea, viz. pairing off the factors, strikes.

Q.18 If $f : [0, \infty) \rightarrow \mathbb{R}$ is a continuous function such that

$$f(x) + \ln 2 \int_0^x f(t) dt = 1, \quad x \geq 0$$

then for all $x \geq 0$,

- (A) $f(x) = e^x \ln 2$ (B) $f(x) = e^{-x} \ln 2$
 (C) $f(x) = 2^x$ (D) $f(x) = \left(\frac{1}{2}\right)^x$

Answer and Comments: (D). The function f is not given to be differentiable. But it is continuous and so the function obtained by integrating it, viz. $\int_0^x f(t)dt$ is differentiable. So, if we rewrite the given condition as

$$f(x) = 1 - \ln 2 \int_0^x f(t)dt \quad (1)$$

then the R.H.S. and hence $f(x)$ is differentiable. Taking derivatives of both the sides of (1), we have

$$f'(x) = -\ln 2 f(x) \quad (2)$$

This is a linear differential equation with general solution

$$f(x) = ce^{-x \ln 2} = c(e^{-\ln 2})^x = c\left(\frac{1}{2}\right)^x \quad (3)$$

where c is an arbitrary constant. Putting $x = 0$ in the given equation, we get $f(0) = 1$ which determines c as 1. Hence the solution is (D).

Conversion to a differential equation and finding its solution are fairly routine. The less trivial part is to show that $f(x)$ is differentiable, even though it is given to be merely continuous. But once again, the MCQ format fails to test it.

Q.19 If $[x]$ denotes the largest integer less than or equal to x , then

$$\left[\left(9 + \sqrt{80}\right)^{20} \right]$$

equals

- (A) $(9 + \sqrt{80})^{20} - (9 - \sqrt{80})^{20}$
 (B) $(9 + \sqrt{80})^{20} - (9 - \sqrt{80})^{20} - 20$
 (C) $(9 + \sqrt{80})^{20} - (9 - \sqrt{80})^{20} - 1$
 (D) $(9 - \sqrt{80})^{20}$

Answer and Comments: (C). When expanded by the binomial theorem,

$$(9 + \sqrt{80})^{20} = \sum_{k=0}^{20} \binom{20}{k} 9^{20-k} 80^{k/2} \quad (1)$$

Terms for which k are irrational because $\sqrt{80}$ and hence all its odd powers are irrational. So $(9 + \sqrt{80})^{20}$ is not an integer. Our task is to find the largest integer, say I less than it.

However, if we consider the conjugate surd $(9 - \sqrt{80})^{20}$, then we get

$$(9 - \sqrt{80})^{20} = \sum_{k=0}^{20} (-1)^k \binom{20}{k} 9^{20-k} 80^{k/2} \quad (2)$$

If we add (1) and (2), the terms for which k is odd get cancelled and we are left with

$$(9 + \sqrt{80})^{20} + (9 - \sqrt{80})^{20} = 2 \sum_{r=0}^{10} \binom{20}{2r} 9^{20-2r} 80^r \quad (3)$$

which is clearly an integer. Of course, it is an integer greater than $(9 + \sqrt{80})^{20}$ because the second term on the L.H.S. is positive. So we have

$$I < (9 + \sqrt{80})^{20} + (9 - \sqrt{80})^{20} \quad (4)$$

But at the same time, the new term we are adding, viz. $(9 - \sqrt{80})^{20}$ is very small because 9 is only slightly bigger than $\sqrt{80}$ and therefore the difference $9 - \sqrt{80}$ is small and its 20-th power would be negligibly small. (To be more precise, $9 - \sqrt{80}$ equals $\frac{1}{9+\sqrt{80}}$ and hence is less than $\frac{1}{9+8}$ which itself is less than 10^{-1} . Hence the term $(9 - \sqrt{80})^{20}$ will be less than 10^{-20} .)

So, (4) can be tightened by

$$(9 + \sqrt{80})^{20} + (9 - \sqrt{80})^{20} - 1 < I < (9 + \sqrt{80})^{20} + (9 - \sqrt{80})^{20} \quad (5)$$

which shows that the integral part of $(9 + \sqrt{80})^{20}$ is $(9 + \sqrt{80})^{20} + (9 - \sqrt{80})^{20} - 1$.

A good problem based on the application of conjugate surds. Similar problems are given in Exercise (6.27).

Q.20 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable one-to-one function. If $f(2) = 2$, $f(3) = -8$ and

$$\int_2^3 f(x) dx = -3$$

then

$$\int_{-8}^2 f^{-1}(x) dx$$

equals

- (A) 25 (B) -25 (C) -31 (D) 31

Answer and Comments: (A). The function $f(x)$ is not given explicitly. Only its values at the two points 2 and 3 are given. The integral of $f(x)$ over the interval $[2, 3]$ is also given. But this does not determine the function uniquely. In fact, there are infinitely many choices for $f(x)$ satisfying these three constraints. Accordingly, there are infinitely many possibilities for the inverse function f^{-1} too. So, on the face of it, the problem will not have a unique answer. The only saving grace is that the end points of the interval of integration of the integral to be evaluated are precisely the values of the integrand at the end points of the interval of integration of the first integral whose value is given.

This sort of reciprocity relationship suggests that the only way to evaluate the second integral is by relating it with the first one in some general way. Such a relationship indeed exists and is given by the following equation where instead of $[2, 3]$ we have a more general interval $[a, b]$. (The other hypotheses remaining the same.)

$$\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a) \quad (1)$$

To prove this, we note that the R.H.S. is the expression $xf(x)|_a^b$, which would occur if we were evaluating the integral over $[a, b]$ of the derivative of $xf(x)$. The derivative of $xf(x)$ is simply $f(x) + xf'(x)$. In other words,

$$\int_a^b f(x)dx + \int_a^b xf'(x)dx = bf(b) - af(a) \quad (2)$$

So the proof of (1) is reduced to showing that

$$\int_{f(a)}^{f(b)} f^{-1}(x)dx = \int_a^b xf'(x)dx \quad (3)$$

To prove this, we substitute $x = f(t)$ in the integral on the L.H.S. This is a valid substitution since the function f is one-to-one and hence by continuity, a strictly monotonic bijection. Then the integral on the L.H.S. becomes $\int_a^b tf'(t)dt$. Here t is a dummy variable of integration and we can change it at will. Replacing it by x gives (3) and hence (1).

It is now trivial to finish the solution. Apply (1) with $a = 2, b = 3, f(a) = 2$ and $f(3) = -8$. Then $f^{-1}(2) = 2$ and $f^{-1}(-8) = 3$. Then we get

$$\int_2^3 f(x)dx + \int_2^{-8} f^{-1}(x)dx = 3f(3) - 2f(2) = -24 - 4 = -28 \quad (4)$$

As the first term on the L.H.S. is given to be -3 , the second one is $-28 + 3 = -25$.

But there is something more. The integral asked is $\int_{-8}^2 f^{-1}(x)dx$ and not $\int_2^{-8} f^{-1}(x)dx$. So the correct answer is not -25 but 25 .

The result (1) is not among the well known results about integrals such as the fundamental theorem of calculus (two forms) or the mean value theorem for integrals. It is more suited for problems. As a result, the problem is rather tricky. But such tricks are worn out if used frequently. See the solution to Q.7 of Paper 2 of ISI 2016 where a similar trick was used.

- Q.21 Three left brackets and three right brackets have to be arranged in such a way that if the brackets are serially counted from the left, then the number of right brackets counted is always less than or equal to the number of left brackets counted. In how many ways can this be done?

(A) 3 (B) 4 (C) 5 (D) 6

Answer and Comments: (C). The description given is precisely that of what is called a balanced arrangement of n pairs of brackets. It is a well known result (see Comment No.6 of Chapter 1) that the number of balanced arrangements of n pairs of brackets is $\frac{1}{n+1} \binom{2n}{n}$. This is called the n -th **Catalan number** and pops up in many combinatorial problems. For $n = 3$, this comes out to be $\frac{1}{4} \times \binom{6}{3} = \frac{20}{4} = 5$.

The result used here is non-trivial and proved by first counting the number of unbalanced pairs. Those who are not familiar with this result can do a hand count. For $n = 1$, there is only one arrangement, viz. $[\]$. For $n = 2$, there are 2, viz. $[\] [\]$ and $[\ [\]]$. For $n = 3$, the first bracket has to be a left bracket. Classify the balanced arrangements according to which of the three right brackets cancels it. If it is the first, there are two possible arrangements after it. If it is the second, there is only one before and only one after it. If the canceling right bracket is the third one, then it is at the end and before it there are two possible arrangements. So, in all, there are $2 + 1 + 2 = 5$ arrangements.

The papersetters have probably deliberately avoided the term balanced arrangements as this will give an automatic clue to those who know Catalan numbers. That would be unfair in a competitive examination.

- Q.22 The limit

$$\lim_{n \rightarrow \infty} \left(2^{-2^{n+1}} + 2^{-2^{n-1}} \right)^{2^{-n}}$$

equals

(A) $=1$ (B) $\frac{1}{\sqrt{2}}$ (C) 0 (D) $\frac{1}{4}$

Answer and Comments: (B). Another complicated limit involving powers. As a first measure of simplification, we note that the variable n occurs only in the powers 2^n and 2^{-n} . Secondly, its integrality is nowhere involved. So we may as well put $t = 2^n$, treat it as a continuous variable and take limit as $t \rightarrow \infty$. So our problem is now reduced to finding L where

$$L = \lim_{t \rightarrow \infty} \left(2^{-2t} + 2^{-t/2} \right)^{1/t} \quad (1)$$

$$= \lim_{t \rightarrow \infty} \left(\frac{1}{2^{2t}} + \frac{1}{2^{t/2}} \right)^{1/t} \quad (2)$$

$$= \lim_{t \rightarrow \infty} \left(\frac{1}{2^{2t}} \right)^{1/t} \left(1 + 2^{3t/2} \right)^{1/t} \quad (3)$$

The first factor is a constant $\frac{1}{4}$. But the second factor is in the indeterminate form ∞^0 . To find its limit, we take its log.

$$\ln \left(1 + 2^{3t/2} \right)^{1/t} = \frac{\ln(1 + 2^{3t/2})}{t} \quad (4)$$

To find the limit of the R.H.S. as $t \rightarrow \infty$, we apply L'ôpital's rule. For this,

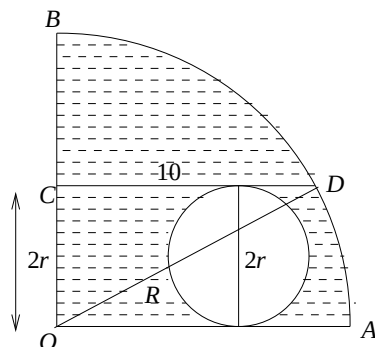
$$\frac{d}{dt} \left(\ln(1 + 2^{3t/2}) \right) = \frac{1}{1 + 2^{3t/2}} \times \ln 2 \times 2^{3t/2} \frac{3}{2} \quad (5)$$

$$= \frac{\frac{3}{2} \ln 2}{1 + 2^{-3t/2}} \quad (6)$$

The numerator and the first term of the denominator are constant. The second term tends to 0 as $t \rightarrow \infty$. Therefore by L'ôpital's rule, (4) tends to $\frac{3}{2} \ln 2$ as $t \rightarrow \infty$. Hence the second factor of the R.H.S. of (3) tends to $e^{\frac{3}{2} \ln 2} = 2^{3/2} = \sqrt{8} = 2\sqrt{2}$. Therefore, we get, finally

$$\begin{aligned} L &= \frac{1}{4} \times 2\sqrt{2} \\ &= \frac{\sqrt{2}}{2} \\ &= \frac{1}{\sqrt{2}} \end{aligned} \quad (7)$$

Q.23 In the following figure, OAB is a quarter-circle. The unshaded region is a circle to which OA and CD are tangents.



If CD is of length 10 and is parallel to OA , then the area of the shaded region in the above figure equals

- (A) 25π (B) 50π (C) 75π (D) 100π

Answer and Comments: (A). Let R and r be the radii of the bigger and the smaller circle respectively. Then the area, say A of the shaded region is

$$A = \frac{\pi}{4}R^2 - \pi r^2 \quad (1)$$

We could find A if we can find the values of R and r . But the data is not sufficient for this. But if we look at (1) carefully, we can recast it as

$$A = \frac{\pi}{4}(R^2 - 4r^2) \quad (2)$$

And, we *can* get the value of $R^2 - 4r^2$ from the right angled triangle OCD if we note that the hypotenuse OD is also a radius of the bigger circle and hence equals R . OA and CD are parallel being the tangents to the smaller circle at the ends of a diameter. Hence OC equals the perpendicular distance between them. The Pythagoras theorem gives $R^2 - 4r^2 = 100$ and so $A = 25\pi$.

The problem is more suited as a puzzle.

Q.24 Consider the function $f : \mathbb{C} \rightarrow \mathbb{C}$ defined by

$$f(a + ib) = e^a(\cos b + i \sin b), \quad a, b \in \mathbb{R}$$

where i is a square root of -1 . Then

- (A) f is one-to-one and onto.
 (B) f is one-to-one but not onto
 (C) f is onto but not one-to-one
 (D) f is neither one-to-one nor onto.

Answer and Comments: (D).

The given definition of f is an elaborate way of saying that the function is simply the exponentiation function, $f(z) = e^z$. It is not onto since it never vanishes. Nor is it one-to-one since $e^{2\pi in} = 1$ for all $n \in \mathbb{N}$.

The purpose of the question is not clear. Once it is realised that the function is the exponential function, all that is needed is recalling some well-known properties of it. Those who can do only computational problems sometimes miss such obvious interpretations. (An anecdote is that a candidate for a lectureship in mathematics was asked to give the Fourier expansion of $\sin x$. The reply was that it will take a long time to calculate it as many integrals will have to be evaluated!)

Q.25 Suppose $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is a non-decreasing function. Consider the following two cases:

Case 1: $f(0) = 2, f(10) = 8$,

Case 2: $f(0) = -2, f(10) = 12$

In which of the above cases it is necessarily true that there exists an n with $f(n) = n$?

- (A) In both cases
- (B) In neither case
- (C) In Case 1, but not necessarily in Case 2.
- (D) In Case 2, but not necessarily in Case 1.

Answer and Comments: (C). It is easy to give a counterexample to disprove the assertion in Case 2. Simply put $f(x) = -2$ for all $x = 1, 2, \dots, 9$. (For points $x < 0$, put $f(x) = x - 100$ and for $x > 10$ put $f(x) = x + 100$. (These are essentially arbitrary choices.) But we claim that the assertion is actually true in Case 1. For, if not then for every integer x either $f(x) > x$ or $f(x) < x$. Colour x red in the first case and blue in the other. Then 0 is red and 10 is blue. So as we scan from 0 to 10, the colour will change from red to blue at some consecutive pair. That is, there will be some x such that $x < f(x)$ and $x + 1 > f(x + 1)$. But $f(x) \leq f(x + 1)$ as f is given to be non-decreasing. Put together, we have

$$x < f(x) \leq f(x + 1) < x + 1 \quad (1)$$

So, the integers $f(x)$ and $f(x + 1)$ (which may be equal) lie in the open interval $(x, x + 1)$. But as x itself is an integer, there are no integers in $(x, x + 1)$, a contradiction. So the assertion is true in Case 1 but not necessarily true in Case 2.

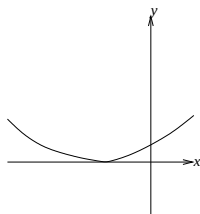
An excellent problem. It is the discrete analogue of the result that if $f(x)$ is continuous on $[a, b]$ and $f(a) > a$ but $f(b) < b$, then there

exists some $c \in (a, b)$ for which $f(c) = c$. (Geometrically, if one end of the graph of a continuous function is below and the other above the line $y = x$, then somewhere it must cross the line $y = x$.) A simple proof is by applying the Intermediate Value Property (IVP) to the function $g(x) = f(x) - x$. Note that no condition like f is monotonically increasing is needed. This is because continuity is a strong condition. For a function of a discrete variable, continuity does not mean much since every function of a discrete variable is continuous. But the hypothesis of monotonicity gives the strength needed.

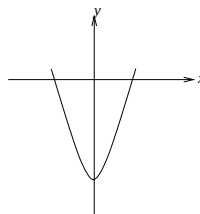
Interestingly, the technique of colouring the points of the domain with two colours and looking for a change of colours can be applied to prove the continuous analogue of the result. For, suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous with $f(a) > a$ and $f(b) < b$. If there is no $c \in (a, b)$ for which $f(c) = c$, colour a point x of $[a, b]$ red or blue according as $f(x) > x$ or $f(x) < x$. Then a is red but b is blue. For a positive integer n , divide the interval $[a, b]$ into n equal parts. The nodes of this partition will be coloured red or blue with the first node (viz. a) red and the last one b blue. So, there are two consecutive nodes, say x_n and $y_n = x_n + \frac{b-a}{n}$ such that x_n is red and y_n blue. (In fact, the number of such pairs must be odd. But that is not needed here. All we need is that there is at least one such pair.) By choosing n sufficiently large, we get that there are arbitrarily close pairs of points with one red and the other blue. This is as far as discrete mathematics takes you. After this the proof can be completed using completeness of real numbers, the most basic result of continuous mathematics.

- Q.26 Suppose that $f(x) = ax^3 + bx^2 + cx + d$ where a, b, c, d are real numbers with $a \neq 0$. The equation $f(x) = 0$ has exactly two distinct real solutions. If $f'(x)$ is the derivative of $f(x)$, then which of the following is a possible graph of $f'(x)$?

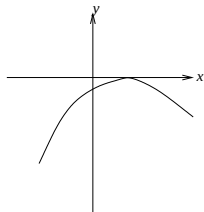
(A)



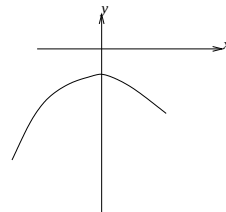
(B)



(C)



(D)



Answer and Comments: (B). Luckily the four possibilities for the real roots of a cubic are listed as the options in Q.4. In the present case, we are given that $f(x) = 0$ has exactly two distinct real solutions. For this to happen, one root, say, α must be a double root and the other, say β distinct from it. But then $f(x)$ is of the form

$$f(x) = a(x - \alpha)^2(x - \beta) \quad (1)$$

Hence

$$f'(x) = 2a(x - \alpha)(x - \beta) + a(x - \alpha)^2 \quad (2)$$

$$= a(x - \alpha)(2(x - \beta) + x - \alpha) \quad (3)$$

$$= a(x - \alpha)(3x - 2\beta - \alpha) \quad (4)$$

Thus $f'(x)$ has two roots, one α and the other different from α (since $2\beta + \alpha \neq 3\alpha$). Among the four possible graphs of $f'(x)$, in (A) and (C) there is a double root, while in (D), there is no root of $f'(x)$. In (B), clearly there are two distinct roots.

The essence of the solution is to show that $f'(x)$ has only two distinct roots. This need not have been done so elaborately as above. We could use the general result that a multiple root of any polynomial is also a root of its derivative. That gives one root, say α of f' . But there is also a root $\beta \neq \alpha$ of the cubic. Hence by Rolle's there is a root of f' between α and β .

Q.27 Suppose $a, b, c \in \mathbb{R}$ and

$$f(x) = ax^2 + bx + c, \quad x \in \mathbb{R}$$

If $0 \leq f(x) \leq (x - 1)^2$ for all x and $f(3) = 2$, then

(A) $a = \frac{1}{2}, b = -1, c = \frac{1}{2}$

(B) $a = \frac{1}{3}, b = -2, c = 0$

(C) $a = \frac{2}{3}, b = -\frac{5}{3}, c = 1$

$$(D) \quad a = \frac{3}{4}, b = -2, c = \frac{5}{4}$$

Answer and Comments: (A). Since $0 \leq f(1) \leq (1-1)^2$ we get that $f(1) = 0$, i.e. 1 is a root of $f(x)$. So the other root, say α , has to be real. If it is different from 1, then on the interval between 1 and α will assume negative values if $a > 0$. But if $a < 0$, then $f(x)$ will assume negative values outside the closed interval with end points 1 and α . In either case, they hypothesis $f(x) \geq 0$ for all x is violated. So we conclude that $\alpha = 1$, i.e. 1 is a double root of $f(x)$, or equivalently $f(x) = k(x-1)^2$ for some constant k . Comparing the leading coefficients, we have $k = a$.

All put together

$$f(x) = a(x-1)^2 = ax^2 - 2ax + a \quad (1)$$

Hence

$$b = -2a, c = a \quad (2)$$

So, b, c are determined as soon as we know. To find a , we use $f(3) = 2$ which gives $a(3-1)^2 = 2$ whence $a = \frac{1}{2}$. From (2), we get $b = -1$ and $c = \frac{1}{2}$.

A simple, but unusual problem on quadratics. The key idea is that a quadratic with different real roots assumes both positive and negative values somewhere.

Q.28 The polynomial $x^{10} + x^2 + 1$ is divisible by

$$\begin{array}{ll} (A) \ x^2 + x + 1 & (B) \ x^2 - x + 1 \\ (C) \ x^2 + 1 & (D) \ x^5 - 1 \end{array}$$

Answer and Comments: (A). We can, of course, apply the long division in each case to find the remainder and get the answer. But obviously, we look for short cuts wherever possible. (C) and (D) can be dismissed easily. Call $x^{10} + x^2 + 1$ as $f(x)$. If $f(x)$ were divisible by $x^2 + 1$, so would $f(x) - (x^2 + 1)$ which equals x^{10} . But a polynomial with no constant term can never be divisible by one whose constant term is non-zero. For (D), note that $f(x) = x^{10} - 1 + x^2 + 2$. But x^{10} is divisible by $x^5 - 1$. So, if $f(x)$ is divisible by $x^5 - 1$ so would be $x^2 + 2$ which is impossible since $x^5 - 1$ has a higher degree.

A short cut of this type is not available for (A) and (B) because no easy multiples of these polynomials can be subtracted from $f(x)$ so that the polynomial left is readily seen to be divisible or not. But fortunately the polynomials in (A) and (B) are quadratics and hence their roots can be easily obtained. Once they are, we can apply the result that whenever a polynomial $f(x)$ is divisible by another polynomial $g(x)$, every root of

$g(x)$ is also a root of $f(x)$. Moreover, the converse is also true if $g(x)$ has no multiple roots. (Actually, even multiple roots are all right, provided the multiplicity of a root of $g(x)$ is at least its multiplicity as a root of $f(x)$.)

We begin with (A). The roots of $x^2 + x + 1$ are ω and ω^2 , where $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2} = e^{2\pi i/3}$ is a complex cube root of unity. As a result, powers of ω recur in cycles of length 3 with $\omega^3 = 1$. This gives

$$f(\omega) = \omega^{10} + \omega^2 + 1 = \omega^2 + \omega + 1 = 0 \quad (1)$$

Similarly,

$$f(\omega^2) = \omega^{20} + \omega^4 + 1 = \omega^2 + \omega + 1 \quad (2)$$

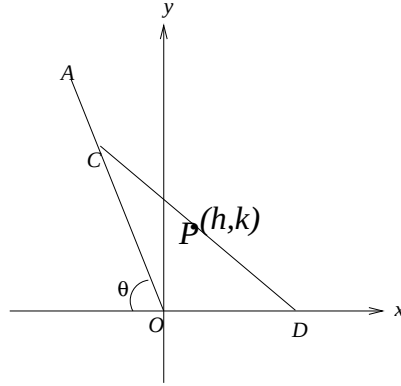
Thus both the roots of $x^2 + x + 1$ are also roots of $f(x)$. Therefore it divides $f(x)$. As only one option is correct, it is now pointless to investigate (B). Still, for the sake of completeness, the roots of $x^2 - x + 1 = 0$ are $\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$ which can be written compactly as $e^{\pm i\pi/3}$. Their powers recur in cycles of length 6 and their cubes equal -1 and not 1 . So,

$$\begin{aligned} f(e^{i\pi/3}) &= e^{10i\pi/3} + e^{i2\pi/3} + 1 \\ &= -e^{i\pi/3} + e^{i2\pi/3} + 1 \end{aligned}$$

whose imaginary part $-\sin \frac{\pi}{3} + \sin \frac{2\pi}{3}$ is non-zero. So without any further checking, $e^{i\pi/3}$ is not a root of $f(x)$. Hence $f(x)$ is not divisible by $x^2 - x + 1$. (We could also have applied this reasoning to (C) because the roots of $x^2 + 1$ are $\pm i$ and $f(i) = i^{10} + i^2 + 1 = -1 - 1 + 1 = -1 \neq 0$. But there a short cut was available anyway.)

In JEE, there are some questions where one or more answer is correct. If this were the case here, each option would have to be dealt with. But as it is given that only one option is correct, and the very first one turns out to be correct, it is a waste of time to check other options. This is one place where the JEE practice is superior. The normal tendency is to try the options in the given order. It would have been better to keep the correct option as (C) or (D).

- Q.29 As in the following figure, the straight line OA lies in the second quadrant of the (x, y) plane and makes an angle θ with the negative half of the x -axis, where $0 < \theta < \frac{\pi}{2}$. The line segment CD of length 1 slides on the (x, y) plane in such a way that C is always on OA and D on the positive side of the x -axis. The locus of the mid-point of CD is



- (A) $x^2 + 4xy \cot \theta + y^2(1 + 4 \cot^2 \theta) = \frac{1}{4}$
 (B) $x^2 + y^2 = \frac{1}{4} + \cot^2 \theta$
 (C) $x^2 + 4xy \cot \theta + y^2 = \frac{1}{4}$
 (D) $x^2 + y^2(1 + 4 \cot^2 \theta) = \frac{1}{4}$

Answer and Comments: (A). Finding the locus of a given point (usually the midpoint) on a ladder moving with one end on the horizontal floor and the other on a vertical wall is a standard problem in coordinate geometry. The present problem is a generalisation of this, in that the wall is not vertical but a tilted one, making an angle θ with the negative axis.

As usual, we let $P = (h, k)$ be midpoint of the segment CD . We need an equation in h and k . That would be easy if the wall were the y -axis. But here we need to introduce an auxiliary variable. Let $\alpha > 0$ be the perpendicular distance of C from the y -axis. Then

$$C = (-\alpha, \alpha \tan \theta) \quad (1)$$

As (h, k) is the midpoint of CD we get

$$D = (2h + \alpha, 2k - \alpha \tan \theta) \quad (2)$$

But D lies on the x -axis. So, $2k - \alpha \tan \theta = 0$, i.e.

$$\alpha = 2k \cot \theta \quad (3)$$

Equating the distance CD with 1 we get another equation

$$(2h + 2\alpha)^2 + \alpha^2 \tan^2 \theta = 1 \quad (4)$$

The locus of $P = (h, k)$ can be obtained by eliminating α between (3) and (4). If we substitute (3) into (4), we get

$$4(h + 2k \cot \theta)^2 + 4k^2 = 1 \quad (5)$$

On expansion,

$$h^2 + 4hk \cot \theta + 4k^2 \cot^2 \theta + k^2 = \frac{1}{4} \quad (6)$$

which gives (A) if we replace h, k by x, y respectively.

A straightforward problem on finding a locus. Art lies only in introducing an auxiliary variable.

Q.30 How many functions $f : \{1, 2, \dots, 10\} \rightarrow \{1, 2, \dots, 2000\}$, which satisfy $f(i+1) - f(i) \geq 20$, for all $1 \leq i \leq 9$, are there?

(A) $10! \binom{1829}{10}$ (B) $11! \binom{1830}{11}$

(C) $\binom{1829}{10}$ (D) $\binom{1830}{11}$

Answer and Comments: (C). The requirement implies, in particular, that the function f is strictly monotonically increasing. So its range, say R , is a set with 10 elements, say $x_1, x_2, \dots, x_{10}$ where $x_i = f(i)$. Conversely, every subset R with 10 elements $x_1, x_2, \dots, x_{10}$ in ascending order determines a unique strictly increasing function from $\{1, 2, \dots, 10\}$. Let A_i be the set of integers strictly between x_i and x_{i+1} for $i = 1, 2, \dots, 9$. Also let $A_0 = \{1, 2, \dots, x_1 - 1\}$ and $A_{10} = \{x_{10} + 1, \dots, 2000\}$. If the restriction $x_{i+1} - x_i \geq 20$ was merely that $x_{i+1} > x_i$, then this would be a familiar problem of putting $2000 - 10 = 1990$ identical balls into 11 distinct boxes A_0 to A_{10} . Each such placement amounts to an arrangement of 1990 identical balls and 10 identical interbox partitions and the number of such arrangements is $\binom{2000}{10}$.

The given restriction on f amounts to saying that each of the boxes A_1 to A_9 has at least 19 balls. So we modify the argument above. First we put 19 balls into each of these 9 boxes. We are now left with $1990 - 171 = 1819$ balls and 10 identical interbox partitions. The number of such arrangements is $\binom{1829}{10}$.

The problem is a good test of paraphrasing the data so that the problem transforms into some more familiar problem. By giving the upper index in the binomial coefficients in all options as either 1829 or 1830, the paper setters have given an implicit hint that the requirement $f(i+1) - f(i) \geq 20$ amounts to saying that the i -th box has at least 19 balls.

ISI BStat-BMath-UGA-2023 Paper 2 with Comments

Q.1 Determine all integers $n > 1$ such that every power of n has an odd number of digits.

Solution and Comments: Trivially 1 has the desired property. But that has been excluded. Excluded are also double digit numbers and, more generally, all numbers which have an even number of digits. If we try the single digit numbers $2, 3, 4, \dots, 9$ we see that some power of them runs into two digits. By the binomial theorem, some powers of $102, 103, \dots$ etc. will have an even number of digits.

But what if we take powers of 100? They are $100, 10000, 100000, \dots$ and more generally $(10)^{2k}$ which has $2k + 1$ digits. Of course, since powers of 10000 are included in those of 100, 10000 is also a successful candidate. More generally, all powers of 100 will work.

We claim that these are the only numbers all whose powers have an odd number of digits. For this we need to find a convenient expression for the number of digits in a number which is such that the number of digits in any power can be calculated easily. As we are using the decimal system, logarithms with base 10 provide such an expression. If k is a fixed positive integer, then note that the smallest and the largest numbers with k digits are 10^k and $10^{k+1} - 1$ respectively. Put differently, an integer n has k digits if and only if $k \leq \log_{10} n < k + 1$, or equivalently, if and only if the integral part of $\log n$ is k . (From now on we suppress the base 10 from the notation.)

It is now easy to come up with a solution. Suppose all powers of n have an odd number of digits. Let

$$\log n = I + f \tag{1}$$

(where I and f are, respectively the integral and the fractional parts of $\log n$.) Then I is even and $0 \leq f < 1$. Taking the k -th power,

$$\log n^k = k \log n = kI + kf \tag{2}$$

Note that kI is an integer. But the integral part of $\log n^k$ may be bigger than kI if kf equals or exceeds 1. Let us see when this happens for the first time as k increases. $f \in [0, 1)$. Unless $f = 0$, $\{f, 2f, 3f, \dots, kf, (k+1)f, \dots\}$ is a strictly increasing sequence, in fact an arithmetic progression. Since $f < 1$, the first time kf equals or exceeds 1, kf cannot exceed 2, because $(k-1)f < 1$ implies $kf < 1 + f < 2$. (This is also obvious intuitively because when the common difference of an A.P. (Here $\{kf\}$) is less than 1, it cannot jump across an interval of length 1 (here the interval $[1, 2]$) in one step.)

So, the first time kf equals or exceeds 1, the integral part of $\log n^k$ is $I + 1$ which is odd (since I is even). But that means that n^k has an even number of digits.

The only exception is when $f = 0$. But that means n is an even power of 10, viz. 10^I .

Hence integers of the form 10^I where I is even are the only numbers which have an odd number of digits in every power.

A simple but imaginative problem. The solution comes easily once the idea of taking logs w.r.t. base 10 and their integral parts strikes. This idea comes after some experimentation. So the problem is a good test of this ability on the part of a candidate.

Q.2 Let $a_0 = \frac{1}{2}$ and a_n be defined inductively by

$$a_n = \sqrt{\frac{1 + a_{n-1}}{2}}, \quad n \geq 1.$$

(a) Show that for $n = 0, 1, 2, \dots$,

$$a_n = \cos \theta_n \text{ for some } 0 < \theta_n < \frac{\pi}{2},$$

and determine θ_n .

(b) Using (a) or otherwise, calculate

$$\lim_{n \rightarrow \infty} 4^n (1 - a_n)$$

Solution and Comments: (a) amounts to showing that $0 < a_n < 1$ for every n . This follows easily by induction on n . a_0 is given to be $\frac{1}{2}$. Assume $0 < a_{n-1} < 1$. Squaring the given relation,

$$a_n^2 = \frac{1 + a_{n-1}}{2} \tag{1}$$

By induction hypothesis, the R.H.S. lies between $\frac{1}{2}$ to 1 and hence its (positive) squareroot lies between $\frac{1}{\sqrt{2}}$ to 1 and hence certainly between 0 and 1.

Now we define $\theta_n = \cos^{-1}(a_n)$. To determine θ_n explicitly in terms of n , we replace a_{n-1} and a_n by $\cos \theta_{n-1}$ and θ_n respectively in (1) and get

$$2 \cos^2 \theta_n - 1 = \cos \theta_{n-1} \tag{2}$$

The L.H.S. equals $\cos 2\theta_n$. Since both θ_{n-1} and $2\theta_n$ lie in the interval $[0, \pi]$ on which the cosine function is injective, we get

$$\theta_n = \frac{\theta_{n-1}}{2} \tag{3}$$

Thus $\{\theta_n\}$ is a geometric progression with initial term θ_0 and common ratio $\frac{1}{2}$. Since $a_0 = \frac{1}{2}$, $\theta_0 = \frac{\pi}{3}$. Therefore

$$\theta_n = \frac{\pi/3}{2^n} \quad (4)$$

which completes the solution to (a).

For (b) we use (a). Although in (a) n was a discrete variable, in finding the limit in (b), we may treat 2^n as a continuous variable t and take the limit as $t \rightarrow \infty$ (as was done in Q.22 of Paper 1). Then the problem reduces to evaluating L where

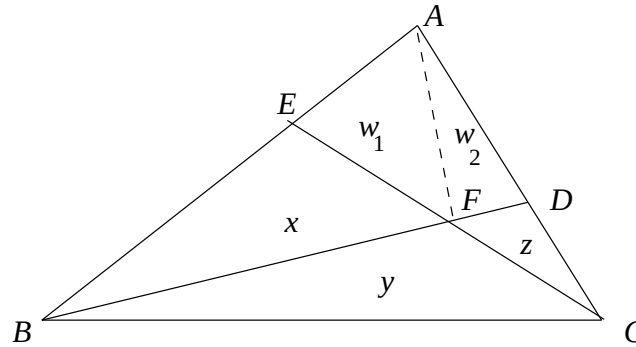
$$L = \lim_{t \rightarrow \infty} t^2 \left(1 - \cos \frac{\pi}{3t}\right) \quad (5)$$

We make one more substitution, $u = \frac{1}{t}$. Then

$$L = \lim_{u \rightarrow 0^+} \frac{1 - \cos(u\pi/3)}{u^2} \quad (6)$$

This can be evaluated using L'Hôpital's rule. Taking first derivatives, we have to find $\lim_{u \rightarrow 0^+} \frac{\frac{\pi}{3} \sin(u\pi/3)}{2u}$. We can apply L'Hôpital's rule again or use the more basic fact that $\frac{\sin \alpha}{\alpha} \rightarrow 1$ as $\alpha \rightarrow 0$. Either way the limit is $\frac{\pi^2}{18}$. So this is also the value of $\lim_{n \rightarrow \infty} 4^n(1 - a_n)$.

Q.3 In a triangle ABC , consider points D and E on AC and AB , respectively, and assume that they do not coincide with any of the vertices A, B, C . If the segments BD and CE intersect at F , consider the areas w, x, y, z of the quadrilateral $AEFD$ and the triangles BEF , BFC , CDF , respectively.



- (a) Prove that $y^2 > xz$.
- (b) Determine w in terms of x, y, z .

Solution and Comments: Both the parts can be solved with a variety of methods. We prefer to prove (a) using vectors and (b) by pure geometry using areas.

Take A as the origin and call $\vec{AB}$ and $\vec{AC}$ (i.e. the position vectors of B and C) as $\vec{b}$ and $\vec{c}$ respectively. Similarly, let $\vec{d}, \vec{e}$ and $\vec{f}$ be the position vectors of D, E, F respectively. Then

$$\vec{d} = \lambda \vec{c} \quad (1)$$

$$\text{and } \vec{e} = \mu \vec{b} \quad (2)$$

for some $\lambda, \mu \in (0, 1)$. Assume F divides BD in the ratio $\frac{\alpha}{1-\alpha}$ and CE in the ratio $\frac{\beta}{1-\beta}$. Then

$$\vec{f} = (1-\alpha)\vec{b} + \alpha\vec{d} = (1-\alpha)\vec{b} + \alpha\lambda\vec{c} \quad (3)$$

But at the same time,

$$\vec{f} = (1-\beta)\vec{c} + \beta\vec{e} = (1-\beta)\vec{c} + \beta\mu\vec{b} \quad (4)$$

As the vectors $\vec{b}, \vec{c}$ are linearly independent, (3) and (4) together give

$$1-\alpha = \beta\mu \quad (5)$$

$$\text{and } 1-\beta = \alpha\lambda \quad (6)$$

which can be solved easily to express α and β in terms of λ and μ

$$\alpha = \frac{1-\mu}{1-\lambda\mu} \quad (7)$$

$$\text{and } \beta = \frac{1-\lambda}{1-\lambda\mu} \quad (8)$$

With this spadework, we are ready to prove (a), i.e. $y^2 > xz$ which is equivalent to proving that $\frac{y}{z} > \frac{x}{y}$. Now,

$$\frac{y}{z} = \frac{BF}{FD} = \frac{\alpha}{1-\alpha} \quad (9)$$

and similarly,

$$\frac{x}{y} = \frac{EF}{FC} = \frac{1-\beta}{\beta} \quad (10)$$

So, (a) is equivalent to proving that

$$\alpha\beta > (1-\alpha)(1-\beta) \quad (11)$$

On expansion, this reduces to proving that

$$\alpha + \beta > 1 \quad (12)$$

For this we apply (7) and (8) which together imply

$$\alpha + \beta = \frac{2 - \lambda - \mu}{1 - \lambda\mu} \quad (13)$$

Hence (a) finally reduces to showing that $2 - \lambda - \mu > 1 - \lambda\mu$ which can be written as $(1 - \lambda)(1 - \mu) > 0$. But that is true because λ and μ are both less than 1 each.

For (b), we draw the line AF . It divides the area w into w_1 and w_2 which are, respectively, the areas of the triangles AEF and AFD . We shall get a system of two equations in w_1, w_2, x, y, z . It can be solved to express each of w_1 and w_2 in terms of x, y, z . Adding, we shall get w in terms of x, y, z .

We note that

$$\frac{AD}{DC} = \frac{\triangle ABD}{\triangle CBD} = \frac{x + w_1 + w_2}{y + z} \quad (14)$$

But at the same time

$$\frac{AD}{DC} = \frac{\triangle AFD}{\triangle CFD} = \frac{w_2}{z} \quad (15)$$

By a property of ratios,

$$\frac{AD}{DC} = \frac{x + w_1 + w_2 - w_2}{y + z - z} = \frac{x + w_1}{y} \quad (16)$$

Hence,

$$\frac{x + w_1}{y} = \frac{w_2}{z} \quad (17)$$

and therefore

$$zw_1 - yw_2 + xz = 0 \quad (18)$$

Interchanging the roles of w_1 and w_2 , and also of x and z , we get a similar equation

$$-yw_1 + xw_2 + zx = 0 \quad (19)$$

(18) and (19) can together be solved as a system of two linear equations in the two unknowns w_1 and w_2 . Skipping the details,

$$w_1 = \frac{xz(y + x)}{y^2 - xz} \quad (20)$$

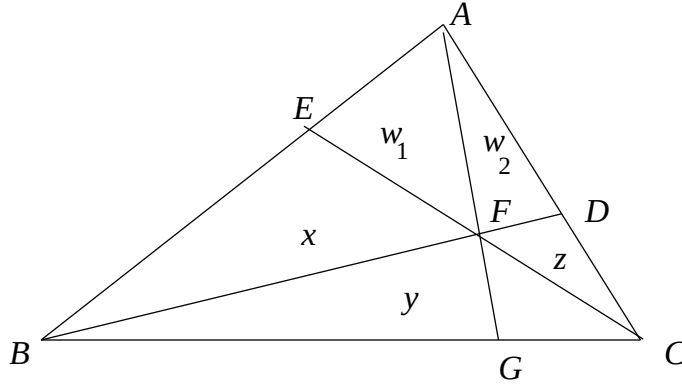
$$\text{and } w_2 = \frac{xz(y + z)}{y^2 - xz} \quad (21)$$

Adding,

$$w = w_1 + w_2 = \frac{zx(x + 2y + z)}{y^2 - xz} \quad (22)$$

which gives an expression for w in terms of x, y and z . It also proves (a) because as the areas are positive, the denominator $y^2 - xz$ is positive, i.e. $y^2 > xz$.

The second solution is shorter and more elegant because it makes use of the powerful concept of areas. The arguments are very similar to those used in the proof of a well-known result in geometry, called **Ceva's theorem**. It is very simple to state. Suppose the line AF when produced meets the side BC at G as shown below.



Then Ceva's theorem says that

$$\frac{AD}{DC} \times \frac{CG}{GB} \times \frac{BE}{EA} = 1 \quad (23)$$

Although this is a result purely about the ratios of lengths of line segments, the standard proof consists of expressing each ratio as a ratio of areas. The ratio $\frac{AD}{DC}$ equals the ratio $\frac{\triangle ABD}{\triangle CBD}$ and also the ratio $\frac{\triangle AFD}{\triangle CFD}$. Hence by the same property of ratios used above, we get

$$\frac{AD}{DC} = \frac{\triangle ABD - \triangle AFD}{\triangle CBD - \triangle CFD} = \frac{\triangle AFB}{\triangle BFC} \quad (24)$$

Similarly,

$$\frac{CG}{GB} = \frac{\triangle AFC}{\triangle BFA} \quad (25)$$

$$\text{and } \frac{BE}{EA} = \frac{\triangle BFC}{\triangle AFC} \quad (26)$$

Ceva's theorem follows by multiplying these three equations.

A good problem on areas in geometry. In (b), it would have been better to ask to prove (22) instead of leaving the expression as open ended, because there could be other expressions for w and some of them very complicated.

- Q.4 Let $n_1, n_2, \dots, n_{51}$ be distinct natural numbers each of which has exactly 2023 positive integer factors. For instance, 2^{2022} has exactly 2023 positive integer factors $1, 2, 2^2, \dots, 2^{2021}, 2^{2022}$. Assume that no prime larger than 11 divides any of the n_i 's. Show that there must be some perfect cube among the n_i 's. You may use the fact that $2023 = 7 \times 17 \times 17$.

Solution and Comments: Let $S = \{n_1, n_2, \dots, n_{51}\}$. Each $n \in S$ has an odd number of divisors. So from the reasoning in Q.17 of Paper 1, each n is a perfect square. Moreover, its only prime divisors are 2, 3, 5, 7 and 11. Hence it is of the form

$$n = 2^a 3^b 5^c 7^d 11^e \quad (1)$$

where a, b, c, d, e are all non-negative even integers (which will change as n varies over S). Every divisor of n is of the form $2^p 3^q 5^r 7^s 11^t$ where p can take values from 0 to a , q can take values from 0 to b and so on. So the number of possible divisors of n are $(a+1)(b+1)(c+1)(d+1)(e+1)$. This is given to be 2023 whose prime factorisation is $7 \times 17 \times 17$. So we get

$$(a+1)(b+1)(c+1)(d+1)(e+1) = 7 \times 17 \times 17 \quad (2)$$

In order that n is a perfect cube, all exponents a, b, c, d, e must be multiples of 3. But we already know that they are all even. So the assertion that S contains a perfect cube is equivalent to saying that there is some $n \in S$ for which a, b, c, d, e are all divisible by 6, or equivalently, each of $a+1, b+1, c+1, d+1$ and $e+1$ takes values which are congruent to 1 modulo 6.

But (2) puts a severe restriction on the possible values of these factors. The possible factorisations of the R.H.S. of (2) lead to the following possible factorisations of the L.H.S.

- (i) (7,17,17,1,1)-type (That means one of the five factors $(a+1), (b+1), (c+1), (d+1)$ and $(e+1)$ is 7, two are 17 and the remaining two are 1).
- (ii) (7, 289, 1,1,1)-type
- (iii) (17,119, 1,1,1)-type
- (iv) (2023, 1, 1, 1,1)-type

In (i), the factor 17 is not congruent to 1 modulo 6, i.e. is not of the form $6k+1$ for any integer k . So if n falls into this type, it cannot be a perfect cube. The same is the case with (iii). However, in (ii) and (iv), all entries

are congruent to 1 modulo 7 and hence any n for which the factors $(a+1)$ to $(e+1)$ are of this type will be a perfect cube, in fact a perfect sixth power.

We are given that $|S| = 51$. Let us see how many of these 51 integers are of the ‘bad’ types, i.e. of types (i) and (iii). This is an elementary counting problem.

In (i), the factor 7 can be picked from the five factors in 5 ways. Once done, from the remaining four factors, the two 17's can be picked in $\binom{4}{2}$, i.e. 6 ways. Hence there are at most 30 elements of S of the type (i).

Similarly there are at most $5 \times 4 = 20$ elements of type (iii). As S has 51 elements, there is at least one $n \in S$ which falls in type (ii) or (iv) and hence is a perfect cube (in fact, a perfect sixth power).

An excellent problem requiring only elementary number theory and combinatorics. The criterion for having an odd number of divisors was already needed in Q.17 of Paper 1. This duplication could have been avoided.

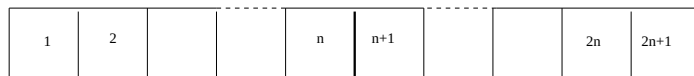
Q.5 There is a rectangular plot of size $1 \times n$. This has to be covered by three types of tiles - red, blue and black. The red tiles are of size 1×1 , the blue tiles are of size 1×1 and the black tiles are of size 1×2 . Let t_n denote the number of ways this can be done. For example, clearly $t_1 = 2$ because we can have either a red or a blue tile. Also, $t_2 = 5$ since we could have tiled the plot as: two red tiles, two blue tiles, a red tile on the left and a blue tile on the right, a blue tile on the left and a red tile on the right, or a single black tile.

- Prove that $t_{2n+1} = t_n(t_{n-1} + t_{n+1})$ for all $n > 1$.
- Prove that $t_n = \sum_{d>0} \binom{n-d}{d} 2^{n-2d}$ for all $n > 0$. Here

$$\binom{m}{r} = \begin{cases} \frac{m!}{r!(m-r)!}, & \text{if } 0 \leq r \leq m \\ 0, & \text{otherwise} \end{cases}$$

for integers m, r .

Solution and Comments: (a) Consider a plot of size $1 \times (2n + 1)$ with $2n + 1$ squares marked from 1 to $2n + 1$ as shown in the diagram below where there is a thick line between the squares marked n and $n + 1$.



This thick line divides the plot into two (unequal) subplots, the left one of size $1 \times n$ and the right one of size $1 \times (n + 1)$.

There are two types of tilings of the entire plot of size $1 \times (2n + 1)$. We can tile the left subplot in any of the t_n ways and simply juxtapose it with any of the t_{n+1} tilings of the right subplot. So, in all, there are $t_n t_{n+1}$ tilings of this type.

But there are also some tilings of the entire plot which are not obtained by freely juxtaposing tilings of the two subplots. These are the tilings in which the n -th and the $(n + 1)$ -th squares are together tiled with a black tile. In this case we can tile the plot with the first $n - 1$ squares in any of t_{n-1} ways and also the $1 \times n$ plot from square $n + 2$ to square $2n + 1$ in any of t_n ways. We then put a 1×2 black tile in between. There are $t_{n-1} t_n$ tilings of this type.

Put together, we get

$$t_{2n+1} = t_n t_{n+1} + t_{n-1} t_n = t_n (t_{n-1} + t_{n+1}) \quad (1)$$

For (b), we classify the t_n valid tilings according to the number, say d , of black tiles. In such a tiling, there are $n - 2d$ non-black tiles. For the time being, we do not distinguish the red tiles from the blue ones. We call both as ‘non-black’.

In a tiling with d black tiles, there are $n - 2d$ non-black tiles. Hence the total number of tiles is $n - 2d + d = n - d$. The number of ways to align d black and $n - 2d$ non-black tiles is simply the binomial coefficient $\binom{n-d}{d}$. But in every such arrangement, the $n - 2d$ non-black tiles can be coloured with red or blue in 2^{n-2d} ways. Hence

$$t_n = \sum_{n \geq 0} \binom{n-d}{d} 2^{n-2d} \quad (2)$$

which proves (b).

A good combinatorial problem requiring careful classification. (b) is very close to the answer to Exercise (7.4.2) in *Foundations of Discrete Mathematics*.

Q.6 Let $\{u_n\}_{n \geq 1}$ be a sequence of real numbers defined as $u_1 = 1$ and

$$u_{n+1} = u_n + \frac{1}{u_n} \text{ for all } n \geq 1$$

Prove that $u_n \leq \frac{3\sqrt{n}}{2}$ for all n .

Solution and Comments: Call $\frac{3\sqrt{n}}{2}$ as v_n . We have to prove that $u_n \leq v_n$ for all n . It is tempting to apply induction on n . But the trouble is that in $u_{n+1} = u_n + \frac{1}{u_n}$ even if $u_n \leq v_n$, there is no guarantee that $\frac{1}{u_n} \leq (v_{n+1} - v_n)$. In fact, an upper bound on u_n gives rise to a lower bound and not an upper bound on the reciprocal $\frac{1}{u_n}$.

Another line of attack is to hope that even though both u_n and v_n increase as n does, the increase in v_n is higher than that in u_n . The trouble is that we are not given u_n explicitly but only as a recurrence relation, which is far from easy to solve. So there is no way to express the increase $u_{n+1} - u_n$ directly in terms of n and compare it with the increase $v_{n+1} - v_n$. In the present case the increase, i.e. $u_{n+1} - u_n$ is $\frac{1}{u_n}$ and this would be less than the increase $v_{n+1} - v_n$ if and only if

$$\begin{aligned} \frac{1}{u_n} &< v_{n+1} - v_n \\ &= \frac{3}{2}(\sqrt{n+1} - \sqrt{n}) \\ &= \frac{3}{2(\sqrt{n+1} + \sqrt{n})} \end{aligned} \quad (1)$$

which is the case if and only if

$$\frac{2(\sqrt{n+1} + \sqrt{n})}{3} \leq u_n \quad (2)$$

So, instead of proving the given inequality we prove the double inequality

$$\frac{2(\sqrt{n+1} + \sqrt{n})}{3} \leq u_n \leq \frac{3\sqrt{n}}{2} \quad (3)$$

by induction on n . Actually, the first inequality is miserably false for $n = 1, 2$. But it is true for $n = 3$ because $u_2 = 2$, $u_3 = \frac{5}{2}$. On the other hand, for $n = 3$ the expression on the left in (3) is $\frac{2(2+\sqrt{3})}{3}$ and so the first inequality reduces to showing that $8 + 4\sqrt{3} \leq 15$, i.e. to $4\sqrt{3} \leq 7$ which is true as we see by squaring both the sides.

So, we prove (3) only for $n \geq 3$. The advantage of proving such a ‘double engine inequality’ is that when dealing with $\frac{1}{u_n}$, we can use the other inequality. We verified the truth of the first inequality just now for $n = 3$. The other inequality also holds for $n = 3$ since $\frac{5}{2} \leq \frac{3\sqrt{3}}{2}$ as we see by squaring both the sides.

Now, for the inductive step, assume (3) holds for some n . To prove it for $n + 1$, we must show that

$$\frac{2(\sqrt{n+2} + \sqrt{n+1})}{3} \leq u_n + \frac{1}{u_n} \leq \frac{3\sqrt{n+1}}{2} \quad (4)$$

From the inductive hypothesis (3), we have

$$u_n + \frac{1}{u_n} \geq \frac{2(\sqrt{n+1} + \sqrt{n})}{3} + \frac{2}{3\sqrt{n}} \quad (5)$$

$$= \frac{2\sqrt{n}(\sqrt{n+1} + \sqrt{n}) + 2}{3\sqrt{n}} \quad (6)$$

$$= \frac{2n + 2 + 2\sqrt{n}\sqrt{n+1}}{3\sqrt{n}} \quad (7)$$

Hence proving the first inequality in (4) reduces to showing that

$$n + 1 + \sqrt{n}\sqrt{n+1} \geq \sqrt{n}(\sqrt{n+1} + \sqrt{n+2}) \quad (8)$$

or, equivalently, to proving that

$$n + 1 \geq \sqrt{n}\sqrt{n+2} \quad (9)$$

which is true as we see by squaring.

We still have to prove the right half of (4). For this we apply the inductive hypothesis (3) with its left half to get an upper bound for $\frac{1}{u_n}$, so that

$$u_n + \frac{1}{u_n} \leq \frac{3\sqrt{n}}{2} + \frac{3}{2(\sqrt{n+1} + \sqrt{n})} \quad (10)$$

$$= \frac{3}{2}(\sqrt{n} + (\sqrt{n+1} - \sqrt{n})) \quad (11)$$

$$= \frac{3\sqrt{n+1}}{2} \quad (12)$$

We have thus proved (4) fully. Hence by induction (3) holds for all $n \geq 3$. Our interest is, of course, only in the right half of (3) which holds true for $n = 1, 2$ by special verification. Hence the given inequality holds for all n .

Admittedly, this is a long proof which was necessitated because u_n was neither given nor could be obtained as an explicit function of n . It turns out that there is a shorter solution which begins by realising that the inequality to be proved is equivalent to showing that $u_n^2 < \frac{9n}{4}$. Squaring the given relation we get

$$u_{n+1}^2 = u_n^2 + 2 + \frac{1}{u_n^2} \quad (13)$$

and recasting this as

$$u_{n+1}^2 - u_n^2 = 2 + \frac{1}{u_n^2} \quad (14)$$

or (for $n \geq 2$,

$$u_n^2 - u_{n-1}^2 = 2 + \frac{1}{u_{n-1}^2} \quad (15)$$

This can be telescoped to get

$$\sum_{k=2}^n u_k^2 - u_{k-1}^2 = 2(n-1) + \sum_{k=2}^n \frac{1}{u_{k-1}^2} \quad (16)$$

i.e.

$$u_n^2 = 2n - 1 + \sum_{r=1}^{n-1} \frac{1}{u_r^2} \quad (17)$$

The terms of the series on the R.H.S. are strictly decreasing (since u_n is strictly increasing) and the largest among them is the first one, viz. 1. This gives $3n - 2$ as an upper bound on u_n^2 . But a more parsimonious estimate is to note that since $u_1 = 1$ and $u_2 = 2$, all terms of the series from the second onwards are at most $\frac{1}{4}$ and decreasing. Hence $2n - 1 + 1 + \frac{(n-2)}{4}$ is a strict upper bound for u_n^2 . But that is less than $\frac{9n}{4}$ and we are done.

- Q.7 (a) Let $n \geq 1$ be an integer. Prove that $X^n + Y^n + Z^n$ can be written as a polynomial with integer coefficients in the variables $\alpha = X + Y + Z$, $\beta = XY + YZ + ZX$ and $\gamma = XYZ$.
- (b) Let $G_n = x^n \sin(nA) + y^n \sin(nB) + z^n \sin(nC)$, where x, y, z, A, B, C are real numbers such that $A + B + C$ is an integral multiple of π . Using (a) or otherwise, show that if $G_1 = G_2 = 0$, then $G_n = 0$ for all positive integers n .

Solution and Comments: We first construct a cubic equation whose roots are X, Y and Z . The simplest such cubic is

$$(u - X)(u - Y)(u - Z) = 0 \quad (1)$$

which when expanded becomes

$$u^3 - \alpha u^2 + \beta u - \gamma = 0 \quad (2)$$

The functions $X + Y + Z$, $XY + YZ + ZX$ and XYZ are called **elementary symmetric functions** of the three variables X, Y, Z . There is a non-trivial theorem of algebra that *any* symmetric polynomial function of X, Y, Z (with integer coefficients) can be expressed as a polynomial in these three elementary symmetric functions. Actually, the number of variables can be more than 3 but then so will be the number of elementary symmetric functions.

The present problem asks to do this for the special case where the function is of the form $X^n + Y^n + Z^n$ for a positive integer n . The standard notation for this function is S_n . We shall prove the assertion directly for $n = 1, 2, 3$ and then by induction for higher n . For notational uniformity $S_0 = X^0 + Y^0 + Z^0 = 1 + 1 + 1 = 3$ (even when one of X, Y, Z may be 0.)

The first case is trivial since $S_1 = X + Y + Z$ is α itself.

For $n = 2$ we write

$$S_2 = X^2 + Y^2 + Z^2 = (X + Y + Z)^2 - 2(XY + YZ + ZX) = \alpha^2 - 2\beta \quad (3)$$

which is also a polynomial in α, β and γ (which is actually missing) and has integral coefficients.

The next case $X^3 + Y^3 + Z^3$ can be handled with the well-known identity

$$X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX) \quad (4)$$

using (3). But we give a proof which is also indicative of the inductive step when we shall prove the assertion for any n . Since X, Y, Z are roots of (2), we have

$$\begin{aligned} X^3 &= \alpha X^2 - \beta X + \gamma \\ Y^3 &= \alpha Y^2 - \beta Y + \gamma \\ \text{and } Z^3 &= \alpha Z^2 - \beta Z + \gamma \end{aligned}$$

Adding,

$$\begin{aligned} S_3 = X^3 + Y^3 + Z^3 &= \alpha(X^2 + Y^2 + Z^2) - \beta(X + Y + Z) + 3\gamma \quad (5) \\ &= \alpha S_2 - \beta S_1 + S_0 \gamma \quad (6) \end{aligned}$$

We can now substitute for S_2, S_1 and S_0 . But that is hardly necessary. The problem merely asks to show that S_3 can be expressed as a polynomial (with integer coefficients) in α, β and γ and not to give a particular expression for S_3 . Since S_2, S_1 and S_0 are already such polynomials, so is S_3 by (6).

Now, for the inductive step, exactly as above we get that for $n \geq 4$

$$S_n = \alpha S_{n-1} - \beta S_{n-2} + \gamma S_{n-3} \quad (7)$$

By the induction hypothesis, the R.H.S. is a polynomial in α, β and γ with integral coefficients and hence so is S_n . This completes the proof of (a).

For (b), we apply (a) for a particular choice of X, Y, Z . Note that nowhere is it needed in (a) that the roots be real. They could as well be complex. We take

$$X = xe^{iA} = x \cos A + ix \sin A \quad (8)$$

$$Y = ye^{iB} = y \cos B + iy \sin B \quad (9)$$

$$\text{and } Z = ze^{iC} = z \cos C + iz \sin C \quad (10)$$

Since $A + B + C$ is of the form $k\pi$ for some integer k ,

$$XYZ = xye^{i(A+B+C)} = xye^{ik\pi} = \pm xyz \quad (11)$$

depending upon whether k is even or odd. The point is that in either case, XYZ is real, i.e. its imaginary part is 0.

We claim that the imaginary part of G_n also vanishes which will prove the result because since x, y, z are all real, for every $n \geq 1$,

$$G_n = x^n \sin nA + y^n \sin nB + z^n \sin nC \quad (12)$$

is precisely the imaginary part of $X^n + Y^n + Z^n$ by DeMoivre's theorem.

Suppose G_1 and G_2 vanish. Then the imaginary parts of $X + Y + Z$ and $X^2 + Y^2 + Z^2$ vanish. But then by (3), the imaginary part of $XY + YZ + ZX$ also vanishes.

Summing up, XYZ is always real, while $X+Y+Z$ and $XY+YZ+ZX$ are real because of vanishing of G_1 and G_2 . By (a), for any n , $X^n + Y^n + Z^n$ is a polynomial with integer coefficients in $X+Y+Z, XY+YZ+ZX$ and XYZ . So $X^n + Y^n + Z^n$ is real and, as already shown, this means that $G_n = 0$.

More generally, there are identities, called **Newton's identities** which express the sums of the n -th powers of the roots of a polynomial in terms of its coefficients. Their use will give a shorter solution but the gain is not substantial, since anyway, our interest is not in an explicit expression for S_n , but only in showing that it can be expressed as a polynomial in the symmetric functions.

A good but lengthy problem in algebra. The stipulation that $A+B+C$ is a multiple of π makes it tempting to think that A, B, C are the angles of a triangle and then look for the millions of trigonometric identities about triangles. But this is a trap and the only way to come out of it is to realise that $A+B+C$ may as well be 2π in which case many of these identities fail. That makes the problem tricky, the correct trick being the choice of the numbers X, Y, Z .

Q.8 Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function which is differentiable on $(0, 1)$. Prove that either f is a linear function $f(x) = ax + b$ or there exists $t \in (0, 1)$ such that $|f(1) - f(0)| < |f'(t)|$.

Solution and Comments: Let $A = (0, f(0))$, $B = (1, f(1))$ and L be the straight line segment joining A and B . Without loss of generality, assume that $f(1) \geq f(0)$. If f is not linear, then the graph of f is not L . Hence there is some point, say, $C = (c, f(c))$ which lies outside L . The slope of L lies between that of AC and of CB . Hence either AC or BC has slope greater than $f(1) - f(0)$. Applying the Lagrange Mean Value Theorem to $(0, c)$ (or to $(c, 1)$ as the case may be), we get a point $t \in (0, 1)$ for which $f'(t) > f(1) - f(0)$.

A very simple problem based on LMVT. Note that f' is not given to be continuous. If it were so, then $f(1) - f(0)$ equals $\int_0^1 f'(x)dx$ which is the average rate of growth of f' . The function f is linear if and only if f' is constant. When it is not linear, we may take t as the point where f' is maximum.

CONCLUDING REMARKS

This year the problems in Paper 2 are much better than those in the recent years. Except possibly for the last problem, every question demands some innovative thinking. The last question could have been dropped to make room for a problem requiring the maximality principle for the solution. (See Q.1 of Paper 2 of ISI 2021.)

But excellent as the problems are, the time constraint may be unfair. It is entirely possible that a candidate can barely solve two or three problems during the test, but after the test solves nearly all in a day or two, while some other candidate solves, say four during the test but afterwards hardly adds one more.

Paper 1 contains some good problems. These are indicated in the respective comments. The best problem is probably Q.19 about an application of conjugate surds. Problem 25 and the last problem (asking for numbers of functions f which satisfy $f(i+1) - f(i) \geq 20$) are close seconds.